Low-Power 8-bit Carry Look Ahead (CLA) Adder

An 8-bit CLA adder designed using Single Phase Adiabatic Dynamic Logic (SPADL) for ultra-low power VLSI applications — built as Minor Project-2 (22UECL605).

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Overview

This project is about designing a faster, lower-power adder for digital circuits. As chips pack in more transistors, power dissipation has become one of the biggest challenges in VLSI design — most of it comes from the constant charging and discharging of capacitors every time a circuit switches. To tackle this, I designed an 8-bit Carry Look Ahead (CLA) adder using an energy-efficient technique called Single Phase Adiabatic Dynamic Logic (SPADL).

Instead of a regular DC supply, SPADL uses a single sinusoidal AC power clock that charges and discharges the load gradually, so a good part of the stored energy can be recovered and reused instead of being wasted as heat. I picked the CLA architecture specifically because it computes carries in advance, making it faster than a simple ripple carry adder, and adders like this are core building blocks in processors and ALUs.

Problem Statement

A number of single-phase adiabatic logic families already exist (SCALD, SCAL, CAL, PAL, QAPG), but they come with real drawbacks:

- They often need extra reference voltages beyond the main supply.
- They add transistor overhead, making the design bigger than necessary.
- Timing control gets complicated.
- Differential signaling increases switching activity.
- Floating output nodes cause capacitive coupling issues.
- All of this eats into the energy efficiency they're supposed to deliver.

To get around these issues, I used a quasi-static Single Phase Adiabatic Dynamic Logic (SPADL) approach and applied it to an 8-bit CLA adder.

Objectives

- Design and implement an 8-bit CLA adder using SPADL.
- Understand how adiabatic logic and energy recovery actually work at the circuit level.
- Build SPADL-based logic gates (inverter, AND, OR, XOR) at the transistor level in Cadence Virtuoso.
- Put together the complete 8-bit CLA architecture using these SPADL gates.
- Simulate everything using Cadence Spectre and check functional correctness.
- Measure power consumption, propagation delay, and Power-Delay Product (PDP).
- Design the physical layout and verify it with DRC and LVS checks.
- Compare the SPADL-based design against a conventional static CMOS CLA adder.

How It Works

The Core Idea

In a normal CMOS circuit, when a capacitor charges and discharges during switching, that energy is simply lost as heat. SPADL instead uses a single sinusoidal AC power clock, so the voltage across the switching devices changes gradually instead of all at once. This lowers the voltage drop during switching, and lets part of the energy be sent back to the supply during the discharge instead of being dissipated.

Two Phases

- **Evaluation phase** — during the rising part of the power clock, the output node charges gradually based on the inputs, while keeping current flow low even under large voltage differences.
- **Recovery phase** — during the falling part of the power clock, the charge stored in the load capacitance flows back to the supply instead of being burned off as heat.

Fitting In the CLA Logic

Each bit in the CLA adder generates two signals — Generate (G) and Propagate (P). These are used to compute carries ahead of time rather than waiting for them to ripple through each stage, which is what makes CLA faster than a ripple carry adder. I built the generate and propagate logic, along with the rest of the gates, using SPADL so the whole adder benefits from the energy recovery.

Design Methodology

The whole design was carried out in Cadence Virtuoso using TSMC 0.18 µm CMOS technology, step by step:

1. Studied how SPADL works — the sinusoidal power clock and the energy recovery mechanism.
2. Designed basic SPADL gates (inverter, AND, OR, XOR) at the transistor level in the schematic editor.
3. Simulated each gate individually with Spectre to check functionality and measure power.
4. Built the Generate (G) and Propagate (P) blocks needed for the CLA using these SPADL gates.
5. Integrated everything into the complete 8-bit CLA architecture.
6. Measured power, delay, and Power-Delay Product (PDP) for the full circuit.
7. Created the physical layout in the Virtuoso layout editor.
8. Ran DRC (Design Rule Check) and LVS (Layout Versus Schematic) to verify the layout.
9. Built a conventional CMOS version of the same CLA adder to compare results against.

Tools Used

- Cadence Virtuoso design environment
- Cadence Spectre simulator
- TSMC 0.18 µm CMOS technology library
- Linux (preferred OS for running the Cadence toolchain)

Where This Can Be Used

- **Low-power microprocessors** — adders are used constantly in ALUs, so power savings add up fast.
- **Portable and battery-powered devices** — phones, tablets, wearables — anywhere battery life matters.
- **Embedded systems** — low-power controllers in IoT and smart devices.
- **DSP systems** — hardware that runs arithmetic operations continuously.
- **Medical electronics** — low-power monitoring devices and implantable electronics.
- **Low-power VLSI research** — as a reference design for energy recovery techniques.

Results

Here's what the design was expected to (and did) show:

- The 8-bit CLA adder worked correctly, verified through transient simulation in Spectre.
- Dynamic switching power dropped noticeably compared to the conventional static CMOS version.

- The sinusoidal power clock produced the expected gradual charging and discharging behaviour.
- Energy efficiency improved thanks to the charge recovery during the falling clock phase.
- Propagation delay stayed within an acceptable range, giving a better overall Power-Delay Product.
- The layout passed DRC and LVS verification.
- Side-by-side comparison confirmed SPADL's power advantage over standard CMOS.

What I'd Add Next

- Extend the design to 16-bit or 32-bit adders to see how the power savings scale.
- Try SPADL on other arithmetic blocks like multipliers or ALUs.
- Explore process-voltage-temperature (PVT) variation analysis for robustness.
- Compare against other single-phase adiabatic families beyond static CMOS.

References

- N.S.S. Reddy, M. Satyam, K.L. Kishore, "Minimization of energy dissipation in glitch free and cascadable adiabatic logic circuits", TENCON 2008, pp. 1–5, Nov. 2008.
- Jianping Hu, Tiefeng Xu, Hong Li, "A lower-power register file based on complementary pass-transistor adiabatic logic", IEICE Trans. on Inf. & Sys., vol. E88-D, No. 7, 2005, pp. 1479–1485.
- D. Maksimovic, V.G. Oklobdziaja, "Integrated Power Clock Generator for Low-Energy Logic", 21st Eur. Solid State Circuit Conf., Lille, France, Sept. 1995.
- M. Chanda, A. Dandapat, H. Rahaman, "Ultra low power sequential circuit implementation by a quasi static single phase adiabatic dynamic logic", TENCON 2009, Singapore, Nov. 2009.